Commerce Platform Product Requirements

Checkout, order and notification business requirements

Document status: CURRENT unless otherwise identified | Enterprise delivery assurance training corpus

1. Purpose and Release Context

Northstar Commerce is preparing a multi-release modernization of its digital commerce platform. This specification is intentionally detailed and includes overlapping business, API, testing, and governance content so that delivery-assurance systems must reason across multiple artifacts rather than rely on one isolated paragraph.

The product requirements define business outcomes, major rules, and release scope for checkout, ordering, and customer notifications. Requirements are referenced by stable identifiers used in the structured dataset.

Area	In Scope	Representative Evidence
Checkout	Cart validation, submission, duplicate prevention	Requirement, API response, test execution
Payments	Authorization, timeout, idempotency	Functional spec, API contract, negative tests
Identity	Authentication, step-up, throttling	Security spec, audit event, test evidence
Release	Traceability, defect thresholds, approvals	Coverage report, execution results, sign-off

1.1 Operational Guidance

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2. Checkout Business Rules

Checkout must validate inventory, price, customer eligibility, and required customer data before payment submission. Duplicate checkout requests using the same business idempotency key must not create multiple orders.

When inventory changes between cart view and checkout submission, the user must receive a stable conflict result and the cart must be refreshed before resubmission.

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3. Order Lifecycle

Order creation should be atomic from the customer perspective. A successfully created order receives a stable order identifier and correlation identifier. Failed submissions must not leave a partially visible order.

Cancellation eligibility depends on order status and fulfillment progress.

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3.3 Delivery Assurance Workflow

Figure: requirement-to-evidence delivery assurance sequence. The visual is an authoritative part of this document and should remain retrievable during ingestion.

The workflow expresses ordering and dependency: requirements are clarified before acceptance criteria are finalized; traceability links requirements to test evidence; execution results and unresolved defects inform the release decision. A system that extracts only nearby paragraphs may lose this sequence.

4. Notification Expectations

Order confirmation and payment-failure notifications must be deduplicated by business event identifier. Transient notification failures may be retried without duplicating customer-visible messages.

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5. Exceptions and Open Questions

Some international address validation rules remain under product review. The system should surface the ambiguity rather than generate a definitive tax or address rule.

Guest checkout retention requirements differ from authenticated customer retention and must be traced to the compliance standard.

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